PKU–Zurich PhD Summer School on Machine Learning for Macroeconomics and Finance

Surrogate-based simulated method of moments in the Brock–Mirman model

Simon Scheidegger Felix Kübler

HEC Lausanne · University of Zurich

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Roadmap of This Lecture

1. **The model.** Brock–Mirman with stochastic TFP, calibration, what is observed.
2. **SMM primer.** Moment condition, weighting matrix, asymptotic distribution.
3. **Why surrogates.** Direct simulation of the SMM objective is too expensive; replace the inner solver with a trained network.
4. **Task 1: Train surrogate** with ϱ as pseudo-state, Euler-residual loss, Gauss–Hermite quadrature.
5. **Task 2: Compute identifying moments** from surrogate-driven simulations.
6. **Task 3: Estimate** $\hat{\varrho}$ via SMM under common random numbers.
7. **Task 4: Joint** (β, ϱ) estimation – identification ridge, over- vs just-identified moment sets.

Learning objectives

By the end of this lecture you should be able to:

- Build and train a neural surrogate with pseudo-state parameters to replace expensive inner-loop solvers
- Compute identifying moments from surrogate-driven simulations under common random numbers
- Implement the SMM estimator using differentiable surrogates and verify interior solutions
- Diagnose parameter identification through SMM-criterion profiles and moment sensitivity
- Extend single-parameter SMM to joint estimation and visualize the identification ridge

Where Session 1 Left Us

The surrogate idea

Train a network $\mathcal{N}(x, \theta)$ *once* on the joint state–parameter space (parameters as **pseudo-states**); afterwards, every counterfactual, moment, or derivative is a cheap forward pass.

Session 1 built the surrogate and previewed its three uses:

1. **Estimate** — point an optimizer at the surrogate's moment predictions. **(today)**
2. *Quantify* — propagate parameter uncertainty through the surrogate. (Session 3, appendix)
3. *Design* — search the policy-parameter space for an optimum. (Session 3)

This session works out the **estimate** operator in full: simulated method of moments on a Brock–Mirman surrogate — first a sharply identified scalar (ϱ), then a partially identified pair (β, ϱ).

Hands-on: `02_01_Structural_Estimation_BM.ipynb` (scalar) and `02_02_Structural_Estimation_BM_Joint.ipynb` (joint).

SMM Primer (1/3): The Moment Condition

Let the data generate T observations $\{y_t\}$. Pick a vector of **moment functions** $g(y) \in \mathbb{R}^q$ such that, under the true parameter $\theta_0 \in \mathbb{R}^p$ $(q \geq p)$,

$$\mathbb{E}\left[g(y_t) - m(\theta_0)\right] = 0,$$

where $m(\theta) = \mathbb{E}_\theta[g(y_t)]$ are the **model-implied moments**. The empirical and simulated counterparts are

$$\widehat{g}_T = \tfrac{1}{T}\sum_{t=1}^{T} g(y_t), \qquad \widehat{m}_S(\theta) = \tfrac{1}{S}\sum_{s=1}^{S} g\left(y_s^{\text{sim}}(\theta)\right).$$

Identification. θ_0 is the unique zero of $\mathbb{E}[g(y_t)] - m(\theta)$ on the parameter space. Just-identified: $q = p$. Over-identified: $q > p$ (testable restrictions).

This lecture. y_t = log output / consumption / capital. g = three autocorrelation and volatility moments. $\theta = \varrho$ (Task 1–3) or $\theta = (\beta, \varrho)$ (Task 4).

SMM Primer (2/3): Estimator and Weighting

The SMM estimator solves

$$\widehat{\theta}_{\mathrm{SMM}} = \arg\min_{\theta} \left[\widehat{g}_T - \widehat{m}_S(\theta)\right]^\top W \left[\widehat{g}_T - \widehat{m}_S(\theta)\right],$$

where W is a positive-definite **weighting matrix**.

Common random numbers (CRN). Hold the random draws $\{\varepsilon_s\}$ fixed across θ. Then $\widehat{m}_S(\theta)$ is a smooth function of θ – simulation noise no longer disrupts the optimizer.

Weighting matrix choice.

- **Identity** $W = I$: easy, but inefficient.
- **Diagonal** $W = \mathrm{diag}(1/\widehat{\sigma}_g^2)$: rescales each moment by its sampling variance.
- **Optimal** $W^\star = \widehat{\Sigma}_g^{-1}$ where $\widehat{\Sigma}_g$ is the long-run covariance of $g(y_t)$. Achieves minimum asymptotic variance.

SMM Primer (3/3): Asymptotic Distribution

Notation follows the script: $M = \partial m(\theta_0)/\partial\theta^\top$ is the moment Jacobian, Σ_m the long-run covariance of the moment vector, and $\Omega = \big(1 + 1/(\tau S)\big)\Sigma_m$ the asymptotic covariance of $\sqrt{T}\hat{g}(\theta_0)$ under S independent simulated panels of length τT relative to the data ($\tau \geq 1$). With S panels of the same length as the data ($\tau = 1$), $\Omega = (1 + 1/S)\Sigma_m$; as $\tau S \to \infty$, $\Omega \to \Sigma_m$.

Under standard regularity conditions (smoothness of m, stationarity, identification), as the data length $T \to \infty$,

$$\sqrt{T}\,(\widehat{\theta}_{\mathrm{SMM}} - \theta_0) \xrightarrow{d} \mathcal{N}\big(0, V(W)\big),$$

$$V(W) \;=\; \big(M^\top WM\big)^{-1} M^\top W\,\Omega\,WM\big(M^\top WM\big)^{-1}.$$

Optimal weighting. With $W = W^\star = \Omega^{-1}$,

$$V(W^\star) \;=\; \big(M^\top \Omega^{-1} M\big)^{-1}.$$

Identification, geometrically. M has full column rank $\Rightarrow$ the moment surface bends sharply along every parameter direction. A flat direction = an *identification ridge* (Task 4 visualizes one).

Why the surrogate? M and m require simulating the model. With a deep surrogate the inner loop is differentiable and three orders of magnitude cheaper.

The Model: Brock–Mirman with Stochastic TFP

Stochastic growth model:

$$\max_{\{C_t\}} \sum_{t=0}^{\infty} \beta^t \mathbb{E}[\ln C_t]\,, \qquad K_{t+1} + C_t = Y_t + (1-\delta)K_t$$

$$Y_t = z_t K_t^{\alpha}, \qquad \log z_{t+1} = \varrho \log z_t + \sigma\, \varepsilon_{t+1}, \qquad \varepsilon_{t+1} \sim \mathcal{N}(0,1).$$

Calibrated parameters

α	δ	σ	β (single-ϱ exercise)
0.36	0.10	0.04	0.96

Note: the textbook closed-form policy $s = \alpha\beta$ requires $\delta = 1$. We use $\delta = 0.10$, so no analytic policy exists and the Euler equation must be solved numerically – the surrogate replaces this expensive inner solve.

Pedagogical sequence:

- **Exercise 1**: estimate $\varrho \in [0.50, 0.99]$ from autocorrelation moments. Sharp 1D identification.
- **Exercise 2**: estimate (β, ϱ) *jointly*. Visualize identification ridge; collapse it with extra moments.

Why ϱ, not β?

- β is famously **poorly identified** from macro moments alone – it shifts the level of the savings rate but every level moment depends on the same scalar.
- In practice, β is calibrated to a target steady-state interest rate (under log utility and full depreciation, $1/\beta = 1 + r$ at SS; with general δ, the SS Euler relation is $1/\beta = 1 + \alpha(k^*)^{\alpha-1} - \delta$).
- ϱ has a **clean direct mapping** to autocorrelation moments: $\mathrm{Corr}(\log Y_t, \log Y_{t-1})$, $\mathrm{Corr}(\Delta \log C_t, \Delta \log C_{t-1})$, and the simulated volatility moment move with persistence.
- This is the textbook example of a **sharply identified** scalar parameter – exactly what we want for teaching the SMM workflow. (With three moments and one parameter the system is technically *over-identified*; the extra moments make the criterion sharper.)

The single-parameter exercise (notebook 02_01) recovers $\hat{\varrho}$ accurately under CRN. The joint exercise (notebook 02_02) shows what happens when an additional, partially-identified parameter is freed.

Task 1: Train Surrogate with ϱ as Pseudo-State

Goal

Train $\mathcal{N}(z_t, K_t, \varrho) \to s_t \in (0, 1)$, with $K_{t+1} = (1-\delta)K_t + Y_t \cdot s_t$.

1. 3-input MLP $(z, K, \varrho) \to s$ with sigmoid output.
2. Euler residual loss with Gauss–Hermite quadrature; ϱ enters the *transition* as well as the policy. Under log utility, $u'(C) = 1/C$; the notebooks minimize the *relative* (scale-free) residual, which shares the zero set of $1 - \beta\, C_t\, \mathbb{E}[\cdot]$ but is better conditioned in FP32:

$$\mathcal{L} = \frac{1}{N} \sum_i \left(\left[\beta\, C_t^{(i)} \sum_j w_j \frac{1 - \delta + \alpha z_{t+1,j}^{(i)} K_{t+1}^{(i)\,\alpha-1}}{C_{t+1,j}^{(i)}} \right]^{-1} - 1 \right)^2$$

3. Two-phase training: (i) uniform sampling on (z, K, ϱ) box, (ii) simulation-based refinement.
4. Validate by plotting $s(z = 1, K, \varrho)$ for several ϱ.

Task 2: Simulate and Compute Identifying Moments

Goal

Generate synthetic "ex-post" data at $\varrho_{\text{true}} = 0.90$ and compute three identifying moments (plus one diagnostic, see below).

Moment	What it captures
$m_1 = \mathrm{Std}(\Delta \log C_t)$	Growth-rate volatility; check numerically
$m_2 = \mathrm{Corr}(\Delta \log C_t, \Delta \log C_{t-1})$	Persistence in consumption growth
$m_3 = \mathrm{Corr}(\log Y_t, \log Y_{t-1})$	Persistence in output (very direct)
$m_0 = \mathrm{mean}(s_t)$ (*diagnostic only*)	Should be flat in ϱ; included to show what *not* to use

Note: $\mathrm{Var}(\log z) = \sigma^2/(1-\varrho^2)$, but $\mathrm{Var}(\Delta \log z) = 2\sigma^2/(1+\varrho)$.

CRN: same burn-in, horizon, initial state, seed for every candidate ϱ.

Task 3: Estimate ϱ via SMM

SMM objective (single-parameter case, $\theta = \varrho$)

$$\hat{\varrho} = \arg\min_{\varrho} \, [m(\varrho) - \hat{m}^{\text{data}}]^\top W \, [m(\varrho) - \hat{m}^{\text{data}}], \qquad W = I.$$

$W = I$ is the unweighted criterion – the cleanest pedagogical default. The efficient choice is $W^\star = \widehat{\Sigma}_g^{-1}$, the inverse long-run covariance of the moment vector (the $\widehat{\Sigma}_g$ of the SMM-primer slides); with $W = I$ the asymptotic variance of $\hat{\varrho}$ is suboptimal but consistency and the interior-solution argument are unchanged.

1. Define one routine $\varrho \mapsto (C_t, I_t, Y_t) \mapsto m(\varrho)$ that reuses the trained surrogate.
2. Evaluate the criterion on a fine ϱ-grid over $[0.50, 0.99]$ (step 0.005, 99 points).
3. Take the grid argmin as $\hat{\varrho}$ – the grid is fine enough that no separate optimizer call is needed.
4. **Verify the solution is interior** ($\hat{\varrho}$ strictly inside the bounds), required for asymptotic theory.

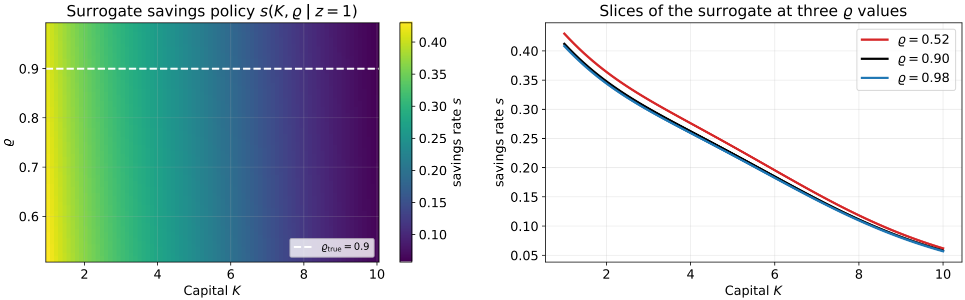
Hint

Re-seed the simulator *to the same seed* on every evaluation, so the candidate parameter is the only thing that changes. Without CRN the criterion is dominated by simulation noise.

Task 4: Diagnose the Fit (Single-Parameter)

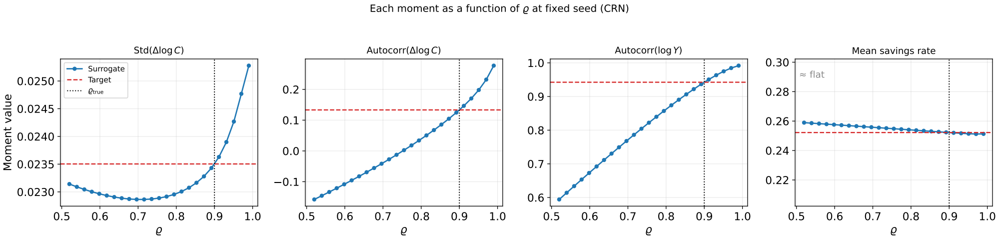
- How close is $\hat{\varrho}$ to $\varrho_{\text{true}} = 0.90$?
- Plot the SMM objective profile and mark both ϱ_{true} and $\hat{\varrho}$.
- Compare matched moments $m(\hat{\varrho})$ versus targets $\hat{m}^{\text{data}}$.
- Plot $s(K, \varrho_{\text{true}})$ vs. $s(K, \hat{\varrho})$ – the curves should overlay almost exactly.
- Trace each candidate moment as a function of ϱ on the trained surrogate – this is the identification picture.

Expected Result: Surrogate Slice $s(K, \varrho \mid z = 1)$



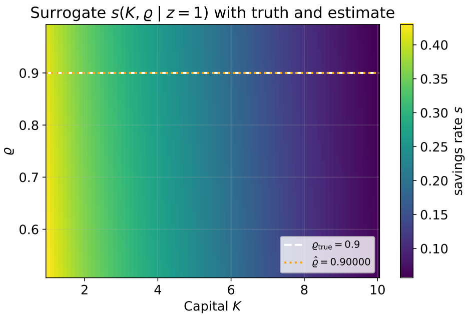
The 3-input network is a smooth function over the entire (K, ϱ) rectangle. The right panel shows three slices; the left panel shows the full surrogate compressed into one heatmap.

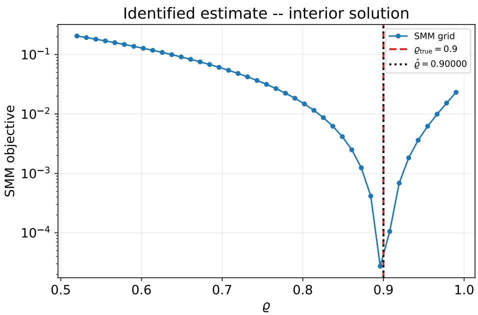
Expected Result: Identifying Variation



All four candidate moments traced as a function of ϱ: the three identifying moments m_1, m_2, m_3 (Task 2 table, leftmost three panels) are monotone or near-monotone in ϱ. The diagnostic moment $m_0 = \mathrm{mean}(s_t)$ (rightmost panel) is essentially flat – correctly diagnosed as uninformative.

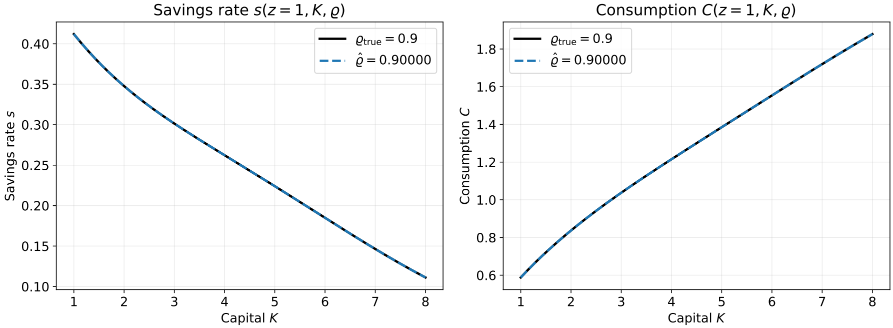
Expected Result: SMM Objective + Estimate





Left: surrogate slice with truth (white dashed) and $\hat{\varrho}$ (orange dotted). Right: log-SMM-objective profile with both. The estimate sits strictly inside the parameter rectangle (**interior** solution).

Expected Result: Policy Match



The estimated policy at $\hat{\varrho}$ overlaps the true policy at $\varrho_{\text{true}} = 0.90$ to plotting accuracy.
Single-parameter SMM with sharp identification recovers the truth accurately under CRN.

Bridge: From Scalar to Joint Estimation

The single-parameter exercise is the cleanest version of the SMM workflow. But two real lessons remain:

- How does the surrogate scale to *multiple* parameters? Same idea: add them as pseudo-states.
- What does the SMM criterion look like when one parameter is **partially identified**?

Notebook `02_02_Structural_Estimation_BM_Joint.ipynb` re-trains the surrogate on the rectangle $\beta \in [0.92, 0.99] \times \varrho \in [0.50, 0.99]$ and estimates both jointly.

Two moment subsets:

- **Weak two-moment** (2 moments): $\{\mathrm{Std}(\Delta \log C), \mathrm{Corr}(\Delta \log C)\}$ – chosen to expose a shallow direction in the criterion.
- **Over-identified** (4 moments): adds mean savings and $\mathrm{Corr}(\log Y)$.

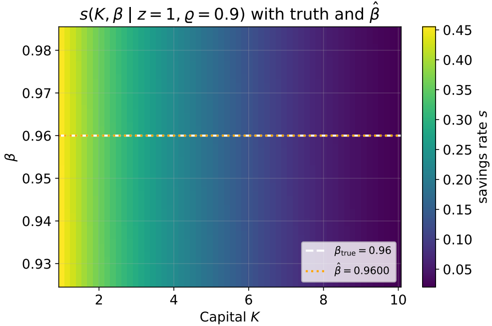
Joint Task: Estimate (β, ϱ)

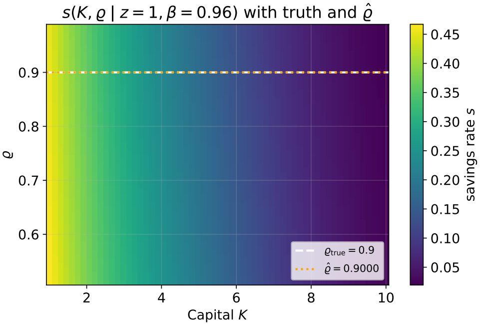
Goal

$\hat{\theta} = \arg\min_{\theta=(\beta,\varrho)} \left[m(\theta) - \hat{m}^{\text{data}}\right]^\top W \left[m(\theta) - \hat{m}^{\text{data}}\right]$ on $\beta \in [0.92, 0.99] \times \varrho \in [0.50, 0.99]$.

1. 4-input MLP $(z, K, \beta, \varrho) \to s$. Train once on the whole rectangle.
2. Forge synthetic data at $(\beta_{\text{true}}, \varrho_{\text{true}}) = (0.96, 0.90)$.
3. Compute the SMM criterion on a 29×50 grid in (β, ϱ) – visualize both moment subsets.
4. Take the grid argmin as $\hat{\theta}$ – gradient-free (no FP32 finite-difference issues); the grid is fine enough that no separate optimizer call is needed.
5. Verify the solution is interior; report distance to truth on each axis.

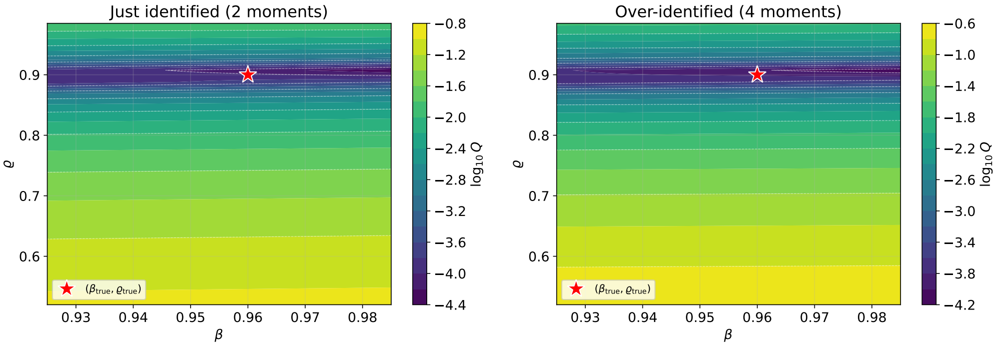
Expected Result: Surrogate Slices on (β, ϱ)





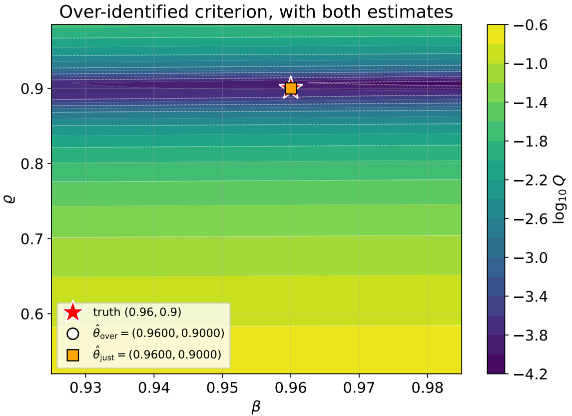
Two slices of the same 4-input network. White dashed = truth, orange dotted = over-identified estimate. The estimate sits strictly **inside** the rectangle on both axes simultaneously.

Expected Result: 2D SMM Criterion



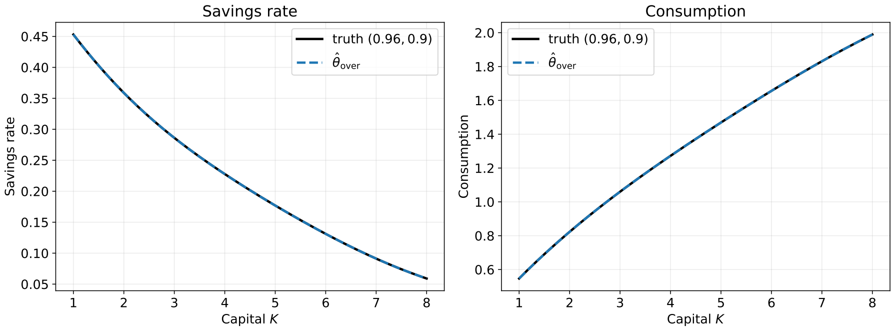
Left: the weak two-moment criterion shows a **ridge** – the partial-identification problem that motivates calibrating β in practice. Right: adding mean savings and output autocorrelation collapses the ridge to a localized minimum.

Expected Result: Estimates on Top of Criterion



In this synthetic CRN run, both estimators recover (β, ϱ) close to the truth. The takeaway is the criterion's *shape*: the just-identified ridge would translate into wide confidence bands on β in real-data applications, while the over-identified criterion has localized curvature on both axes.

Joint Policy Match



Savings rate and consumption at the over-identified estimate overlay the truth almost exactly. The 4-input surrogate, trained *once*, supports millions of cheap moment evaluations during the optimizer's line-search.

Wrap-Up

What the two notebooks teach together:

- A neural surrogate with parameters as pseudo-states **decouples** the cost of solving the structural model from the cost of moment-matching. Train once, use it inside any optimizer.
- For a well-identified parameter (ϱ), SMM under CRN gives a sharp 1D identification picture.
- For **joint** estimation, the contour plot makes the partial-identification ridge visible. Adding moments collapses it.
- The **interior-solution** check is part of the standard pipeline, needed for the asymptotic distribution theory of the SMM estimator.
- **Synthetic-data caveat.** Exact recovery in both notebooks reflects self-consistency: the data are generated by the same surrogate at the true parameter, and every candidate evaluation reuses the same shock path. Real-data SMM does not produce a zero criterion.

Formal identification machinery (moment Jacobian $M = \partial m / \partial \theta$, rank condition, J-statistic, weak-identification diagnostics): see lecture script Sec. 10.4 (Practical Considerations).

Production pipeline: Chen, Didisheim & Scheidegger (2026, *J. Financial Economics*), exactly this surrogate-and-SMM logic at scale, applied to finance.

Next up (Session 3): the same surrogate machinery answers *design* questions — constrained optimal carbon taxes — and, in the appendix of that deck, *quantify* questions (uncertainty quantification of the social cost of carbon).